# OPTIMIZED ANFIS RUNTIME ARCHITECTURE & PIPELINE FLOW (C = 64)

## Stage 1: Memory Access Ingestion & Fast Hit Path

**Incoming Access**
$x_t$ at timestamp $t_{\text{curr}}$
(Address & Op)

**Hit in Cache $\mathcal{C}$?**

YES

**Fast-Path Hit (~5.15 ns)**
- Update $t_{\text{last}} \leftarrow t_{\text{curr}}$
- $F \leftarrow \min(F + 1, 20)$
- Deliver Data to CPU

Enter Eviction Pipeline

NO (Miss)

**Direct Insertion**
- Allocate free slot
- $t_{\text{arrival}} \leftarrow t_{\text{curr}}$
- Fetch from DRAM/SSD

NO

**Cache Full? ($|\mathcal{C}| = 64$)**

Cache Full

## Stage 2: Ghost-Cache Online Learning (Widrow-Hoff LMS)

**Key $x_t$ in Ghost $\mathcal{B}$?**

YES (Ghost Hit)

**Instantaneous Online LMS Step**
- Target $y^* = 1.0$ (Premature Eviction)
- Compute Error: $e_t = 1.0 - \text{Score}_{\text{ghost}}$
- Widrow-Hoff Delta Rule Update:
$c_i^{(t+1)} = \text{clamp}((1-\lambda)c_i^{(t)} + \lambda c_i^{(0)} + \eta e_t \bar{w}_i)$
- 19 CPU Instructions (~15 ns, 0.0000 s CPU)

## Stage 3: Out-of-Band CART Meta-Controller (Amortized N = 250)

**5-D Telemetry Feature Extractor**
Sampled every $N = 250$ reqs:
- $f_1 = U/N$ (Expansion)
- $f_2 = H$ (Shannon Entropy)
- $f_3 = \text{Corr}$ (Autocorr)
- $f_4 = G$ (Gini Skew)
- $f_5 = S$ (Stride Index)

Normalized Feature: $\mathbf{f} \in [0, 1]^5$

**CART Decision Tree Classifier**
Trained on balanced 4-regime bank:
- If $H > 0.94$ and $u \le 0.50$ (Cyclic Loop):
→ Activate Recency Inversion $(1 - R)$
- If $H \le 0.94$ and $\rho \le 0.073$ (Scan):
→ Frequency Shielding $(1.5F, 0.5R)$

Latency: 1.40 $\mu$s (Amortized 5.6 ns/req)

Dynamic Hedge Modulation

## Stage 4: L1-Cache-Resident Neuro-Fuzzy Inference Engine & Tournament Eviction (24[illegible] ns)

**Eviction Tournament (K = 12 Candidates)**
When $|\mathcal{C}| = 64$ is full:
- Partial Fisher-Yates shuffle
- Draw $K = 12$ candidates
- Bounds scan to $O(1)$
- Avoids full $O(C)$ sweeps
- Read packed metadata:
$\{t_{\text{last}}, \text{freq}, t_{\text{arrival}}\}$

**Input Normalization & Tri-Input Synthesis**
For each candidate $k \in \mathcal{K}$:
- $R = \text{clip}(1 - \frac{\Delta t}{4C}, 0, 1)$
- $F = \text{clip}(\frac{\text{freq}}{20}, 0, 1)$
- $A = \text{clip}(1 - \frac{\Delta t_{\text{arr}}}{4C}, 0, 1)$
- Composite Coordinates:
$R_{\text{eff}} = (0.25R + 0.10A)/0.35$
$F_{\text{eff}} = F$

**Layer 1: Fuzzification via Hardware L1 LUT (6 KB)**
Precomputed Gaussian curves:
- Table Size: 6.0 KB (L1D Fit)
- Discretization: 256 points
- Quantized Integer Index:
$i_R = \lfloor 255 R_{\text{eff}} \rfloor$, $i_F = \lfloor 255 F_{\text{eff}} \rfloor$
- Direct $O(1)$ L1 Read:
$\mu_L, \mu_M, \mu_H$ in 1-2 ns

**Layer 2-5: Rule Firing & Defuzzification Engine**
8-Rule Takagi-Sugeno Model:
- L2 Product T-Norm:
$w_i = \mu_{R,i} \times \mu_{F,i}$
- L3 Firing Normalization:
$\bar{w}_i = w_i / \sum_{j=0}^{7} w_j$
- L4 Consequent Priorities: $c_i$
- L5 Zero-Order Centroid:
$\text{Score} = \sum_{i=0}^{7} \bar{w}_i c_i$

**Victim Selection & Cache Replacement**
Tournament Decision:
- Victim $= \arg\min_k (\text{Score}_k)$
- Evict victim from $\mathcal{C}$
- Record tag & score to FIFO Ghost Cache $\mathcal{B}$
- Insert incoming block $x_t$
- Line-Rate Latency: 240.3 ns